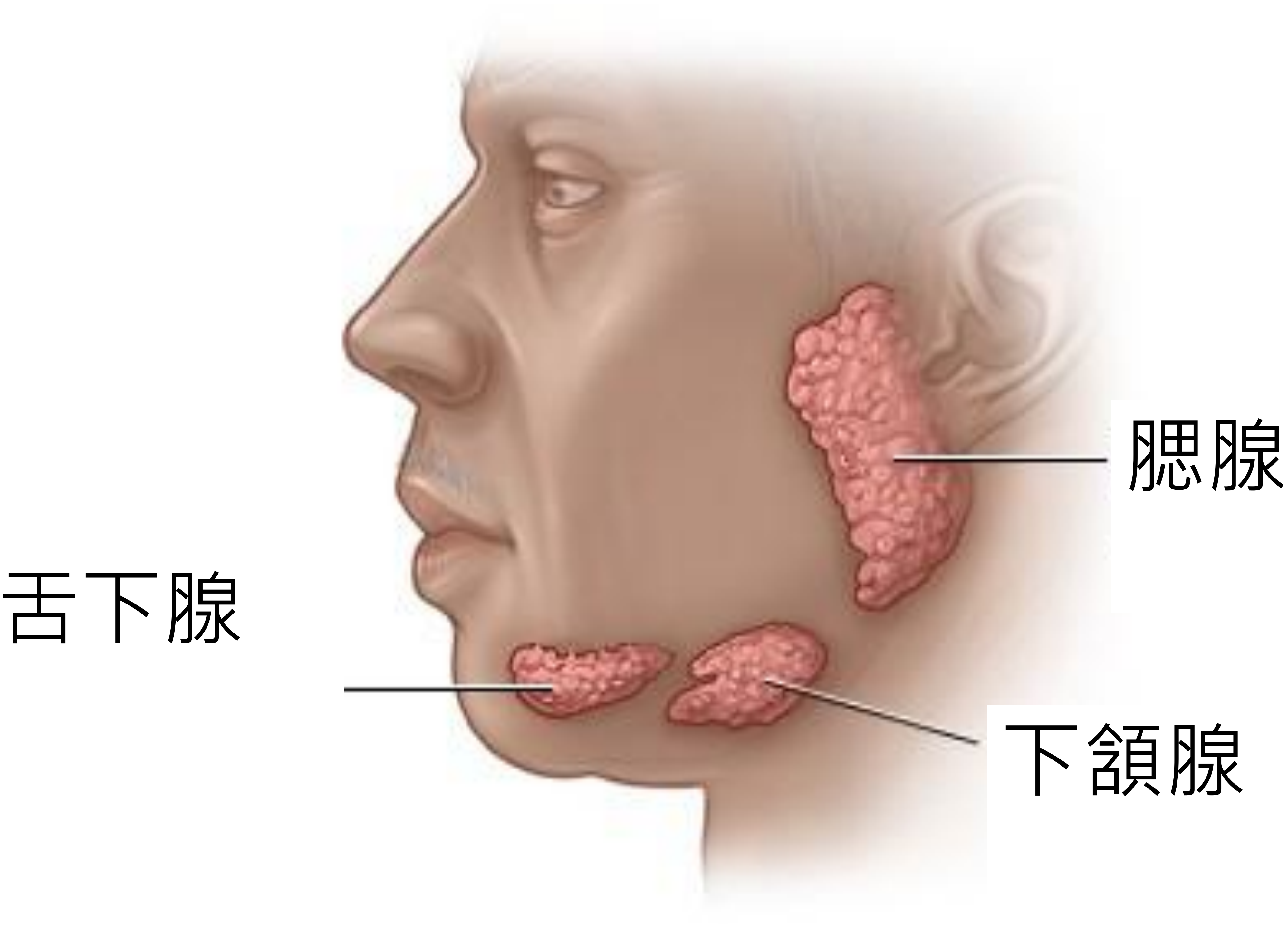
Salivary glands ultrasonography (SGUS)

VS 曾吉騰

Department of Pediatrics
Chang Gung Memorial Hospital, Keelung

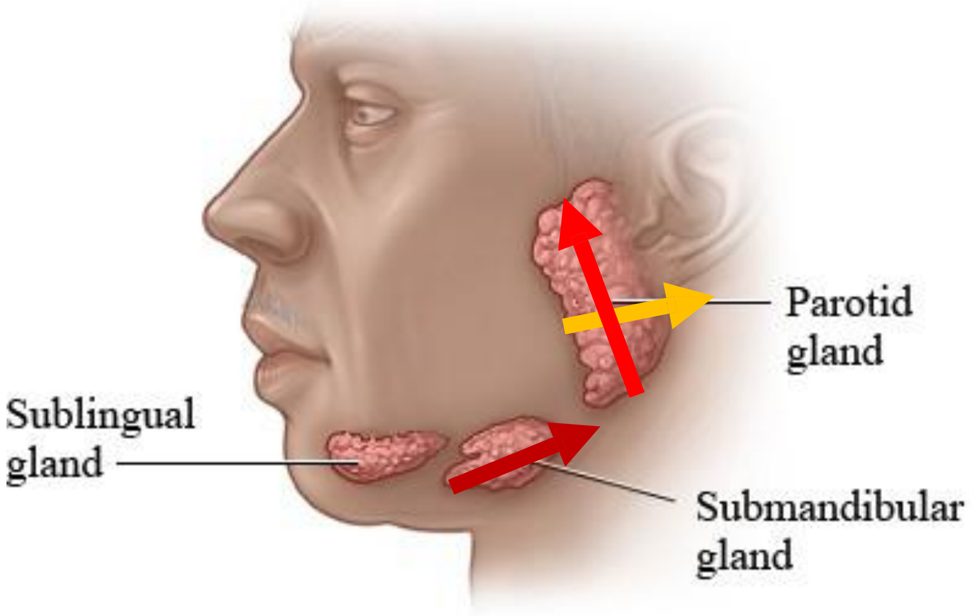

腮腺
舌下腺
下頜腺

Basic concepts

- Most study using 8-13MHz for salivary gland detection, linear probe. 10MHz is the most common one.
- Many surrounding structures
  - Parotid gland: muscle, mandible bone
  - Submandibular gland: large vessels, mandible bone
- Thyroid gland / masseter muscle as reference

How to Perform SGU

- Transverse and longitudinal view of parotid glands.
- Longitudinal view of submandibular glands.



Score system

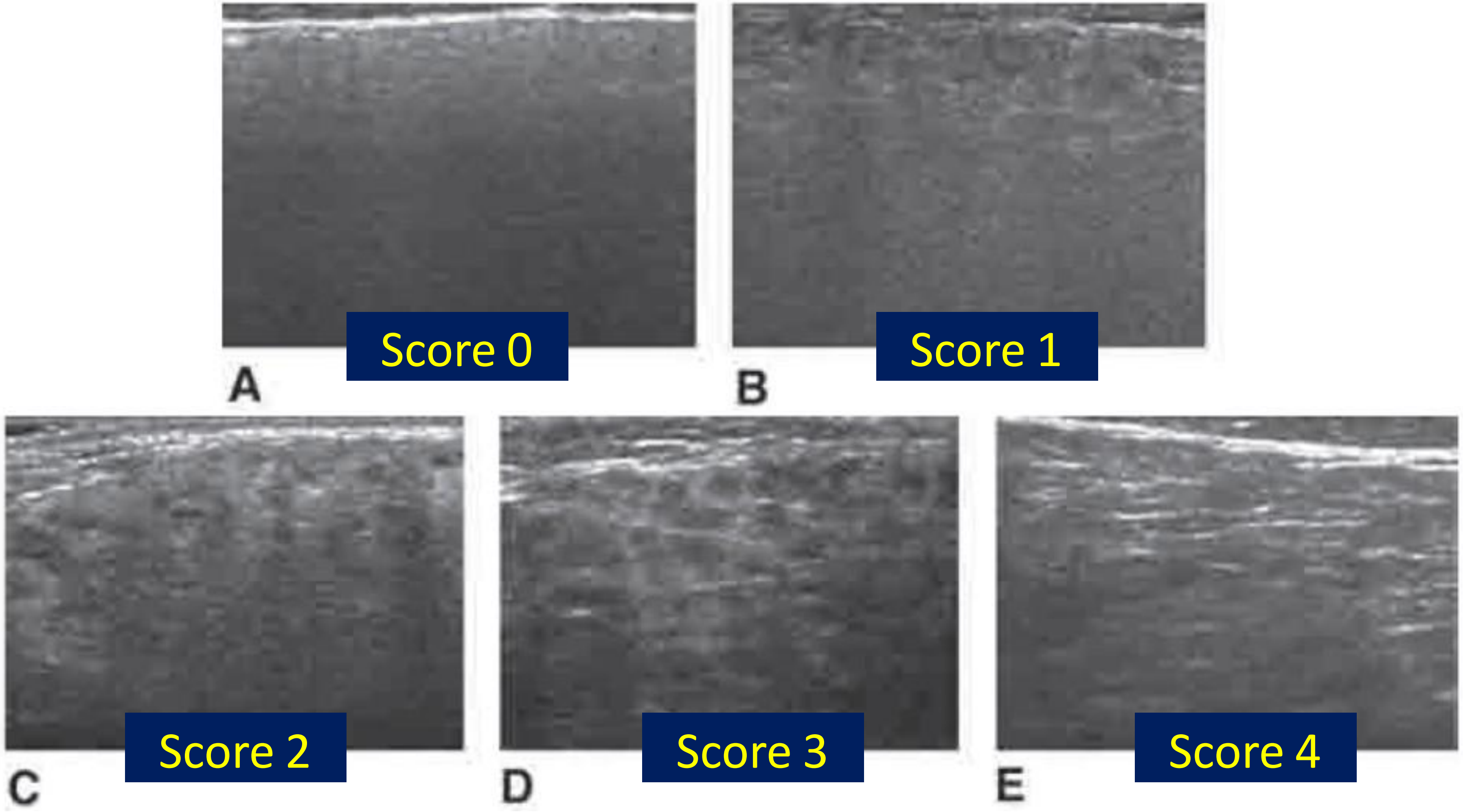
- Several parameters could be included
  - Size: increased / normal / decreased
    - Parotid 20 ± 3mm, submandibular 13 ± 2mm
  - Echogenisity: increased / decreased
  - Inhomogeneity: score 0-3 / score 0-4
  - Focal change: hypo / hyper- echogenic reflections
  - Posterior border: ill-defined / not visible

Table I. US grading score for the evaluation of salivay glands proposed by Salaffi F *et al.* (67).

Grade 0	Normal glands
Grade 1	Regular contour, small hypoechoic spots/areas, without echogenic bands, regular or increased glandular volume (mean values 20 + 3 mm for the parotids and 13 + 2 mm for the submandibular glands) and ill-defined posterior glandular border (definite echogenic border with respect to the neighbouring structures)
Grade 2	Regular contour, evident multiple scattered hypoechogenic areas usually of variable size (<2 mm) and not uniformly distributed, without echogenic bands, regular or increased glandular volume and ill-defined posterior glandular border
Grade 3	Irregular contour, multiple large circumscribed or confluent hypoechogenic areas (2–6 mm) and/or multiple cysts, with echogenic bands, regular or decreased glandular volume and no visible posterior glandular border
Grade 4	Irregular contour, multiple large circumscribed or confluent hypoechogenic areas (>6 mm), and/or multiple cysts or multiple calcifications, with echogenic bands, resulting in severe damage to the glandular architecture, decreased glandular volume and posterior glandular border not visible

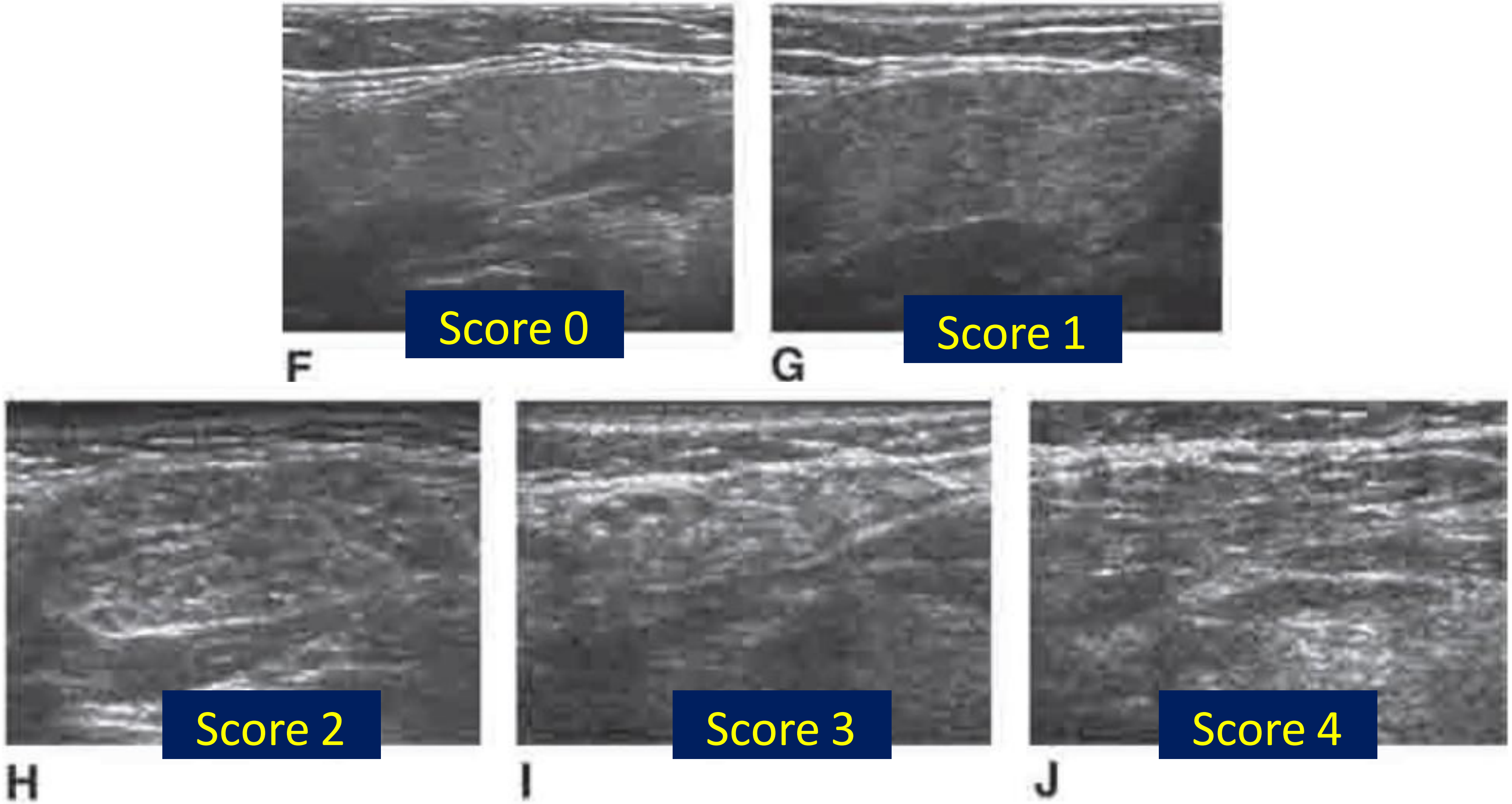
	Score 1	Score 2	Score 3	Score 4
Contour	Regular	Regular	Irregular	Irregular
Volume	Regular or increased	Regular or increased	Regular or decreased	Decreased
Echogenic band	No	No	Yes	Yes
Posterior border	Ill-defined	Ill-defined	Not visible	Not visible
Inhomogeneity	Small hypoechoic areas	< 2mm	2-6mm	>6mm
Others				Loss of normal architecture

Score 0-4 (Parotid gland)



Takagi Y, et al. Ann Rheum Dis. 2010 Jul;69(7):1321-4.

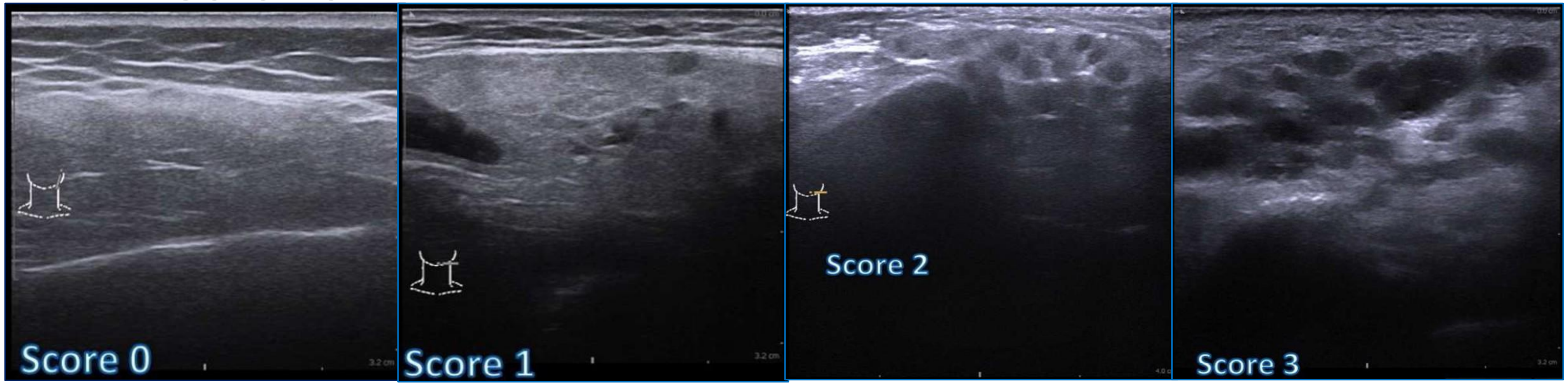
Score 0-4 (Submandibular gland)



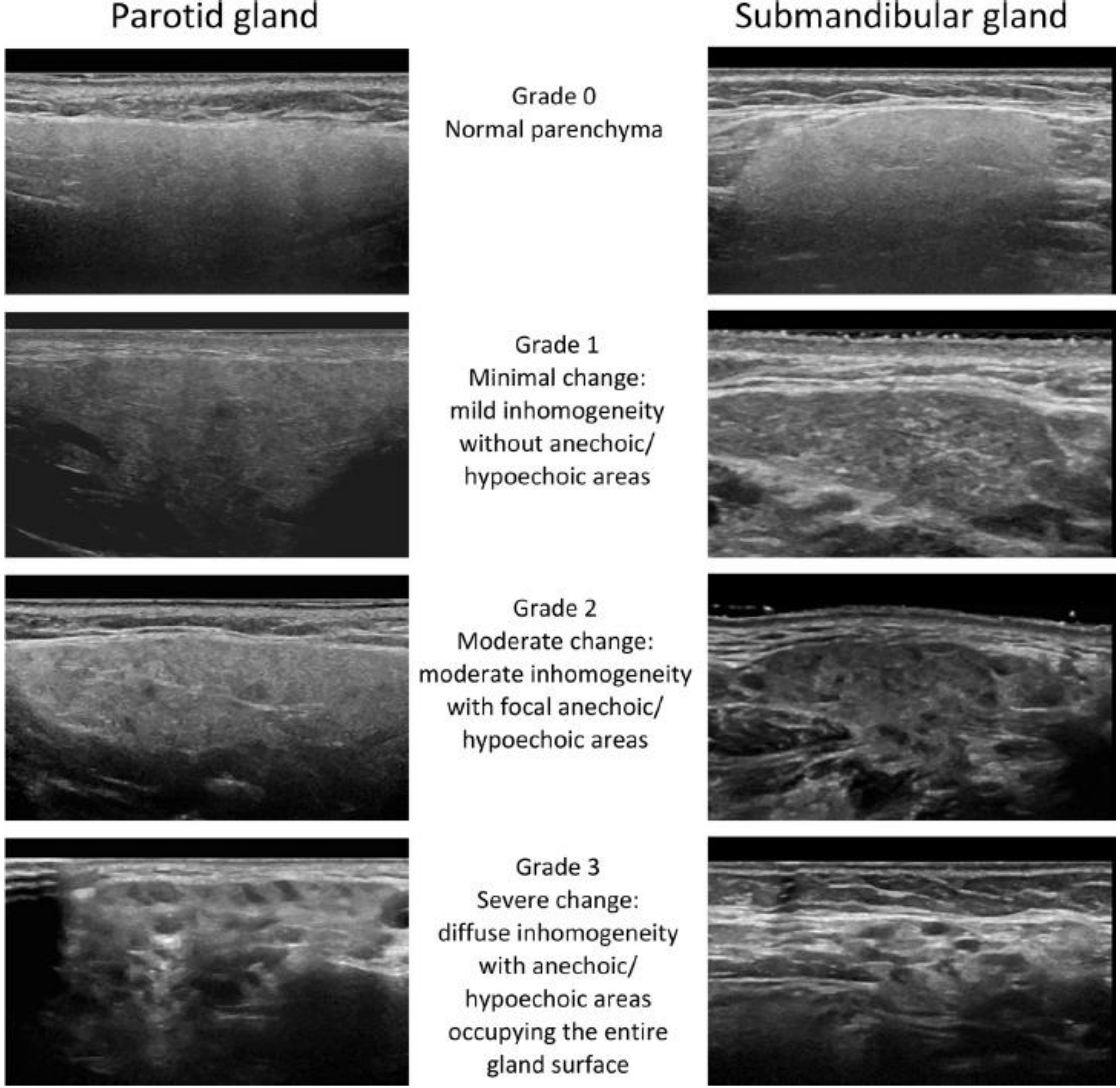
Takagi Y, et al. Ann Rheum Dis. 2010 Jul;69(7):1321-4.

Score system: 0-3 (OMERACT)

- Score 0: normal
- Score 1: mildly inhomogeneous
- Score 2: several rounded hypoechoic lesions
- Score 3: numerous or confluent hypoechoic lesions



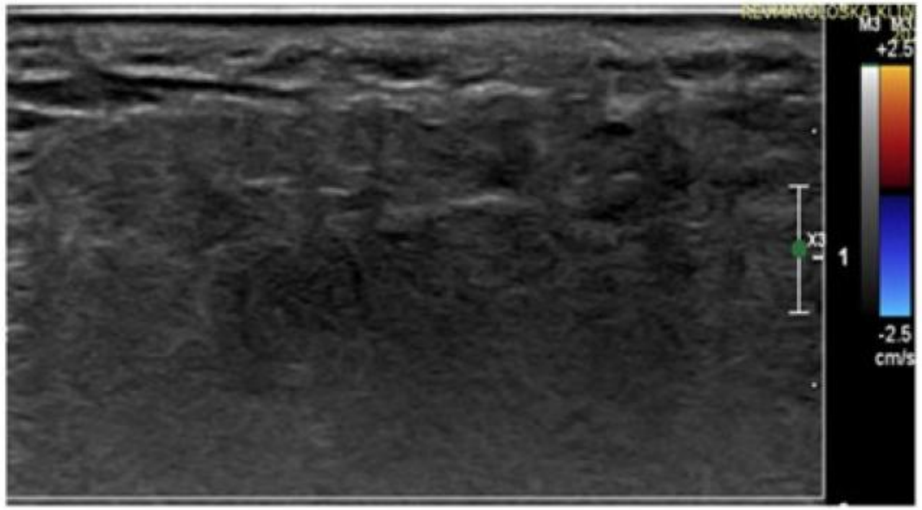
Theander E, Mandl T. Arthritis Care Res (Hoboken). 2013 Dec 10.



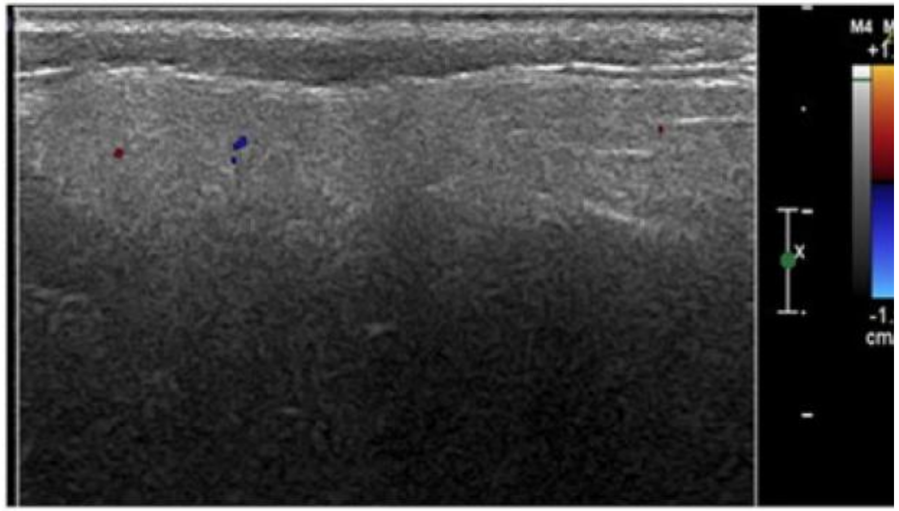
RMD Open. 2021 Apr;7:e001516.

Color Doppler ?

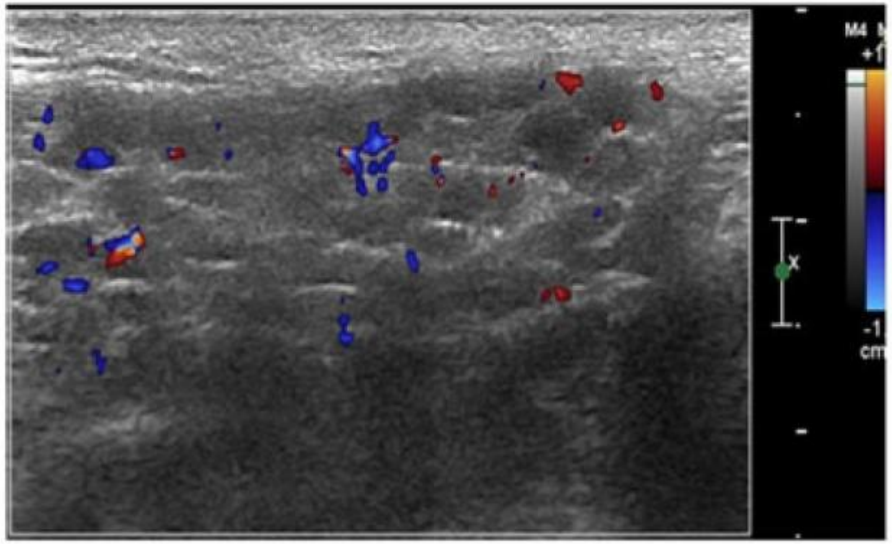
Grade 0



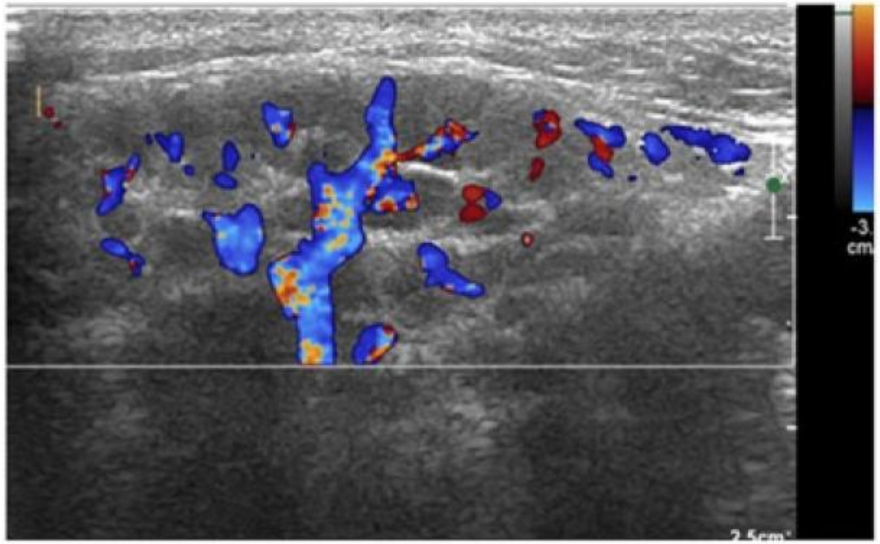
Grade 1



Grade 2

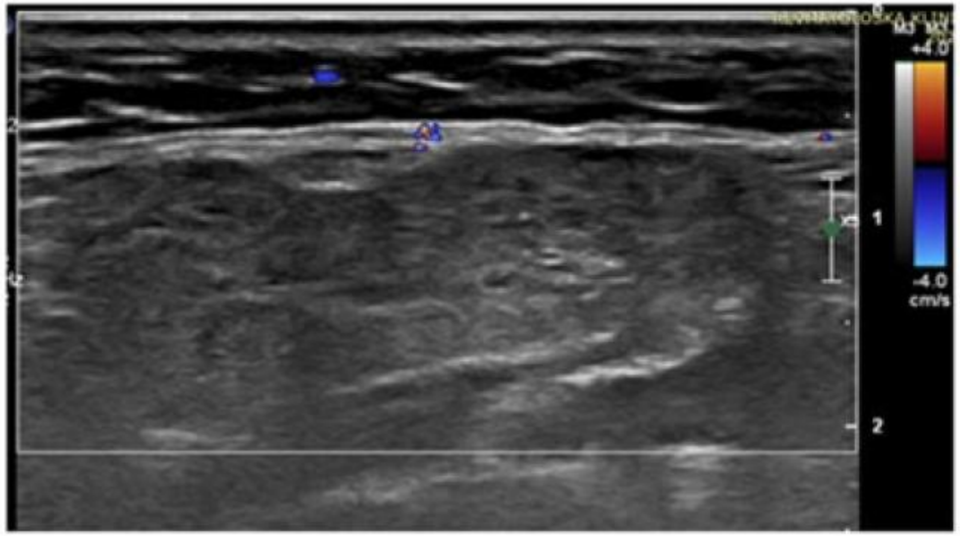


Grade 3

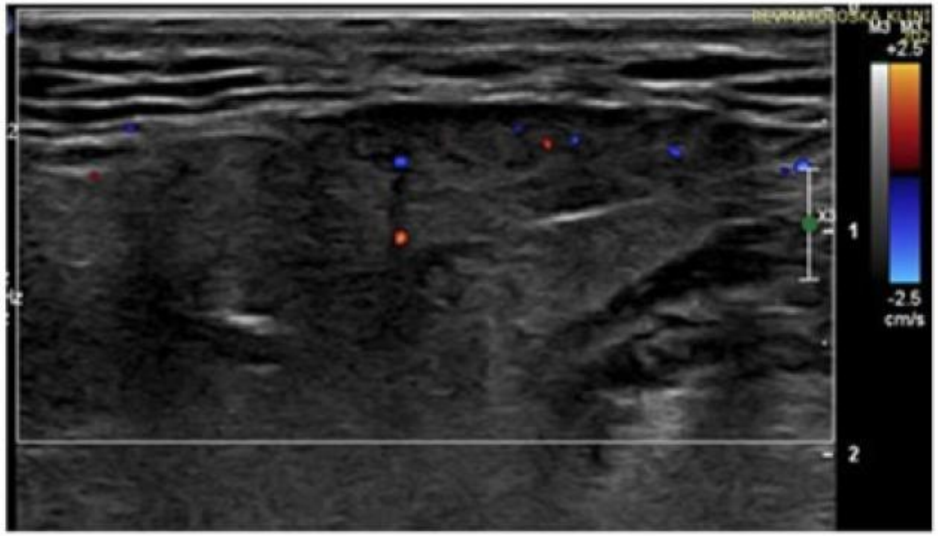


Rheumatology (Oxford) 2022, 61(8), 3341–3350.

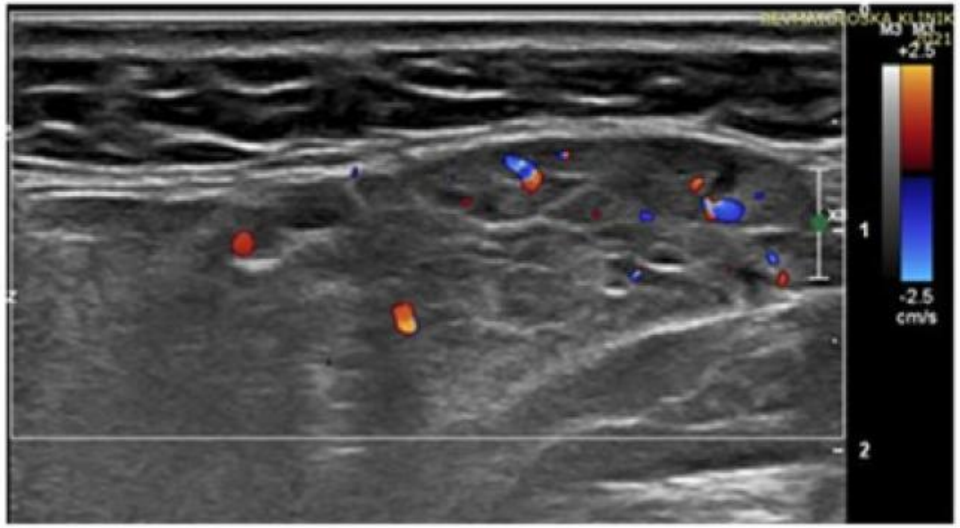
Grade 0



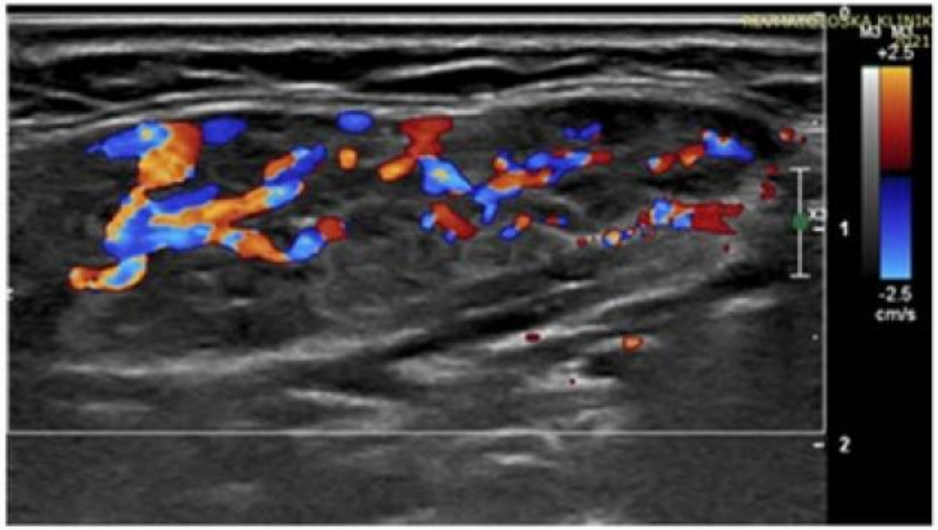
Grade 1



Grade 2

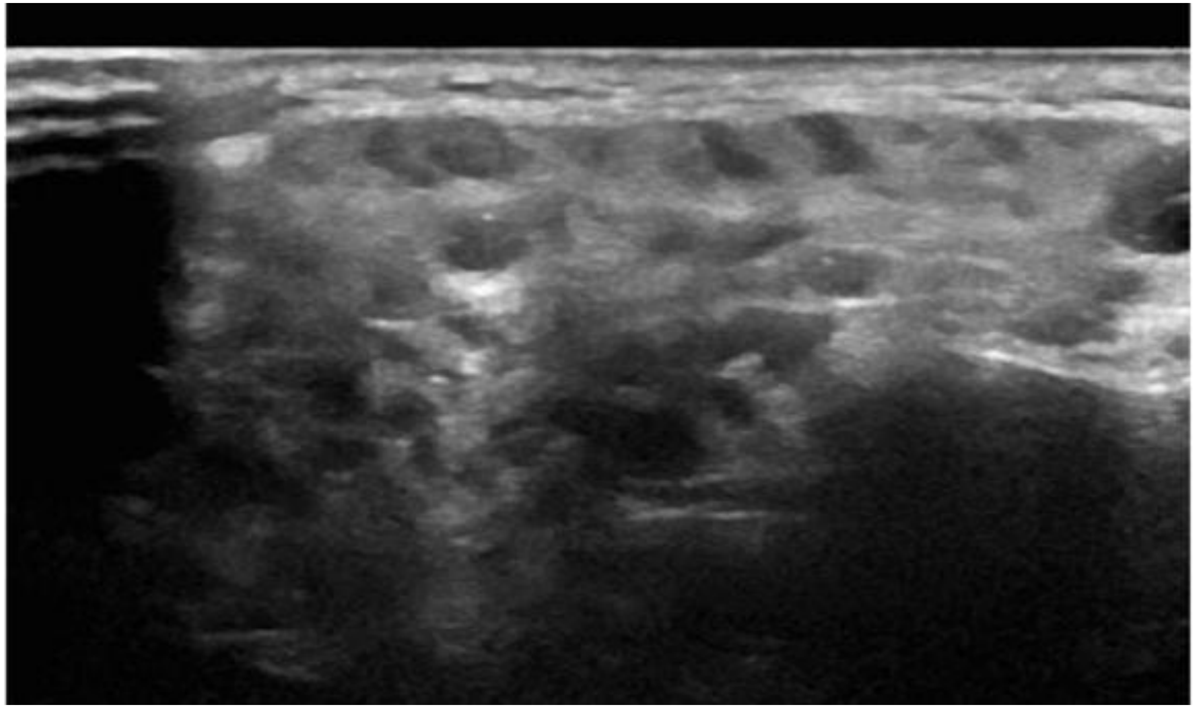


Grade 3

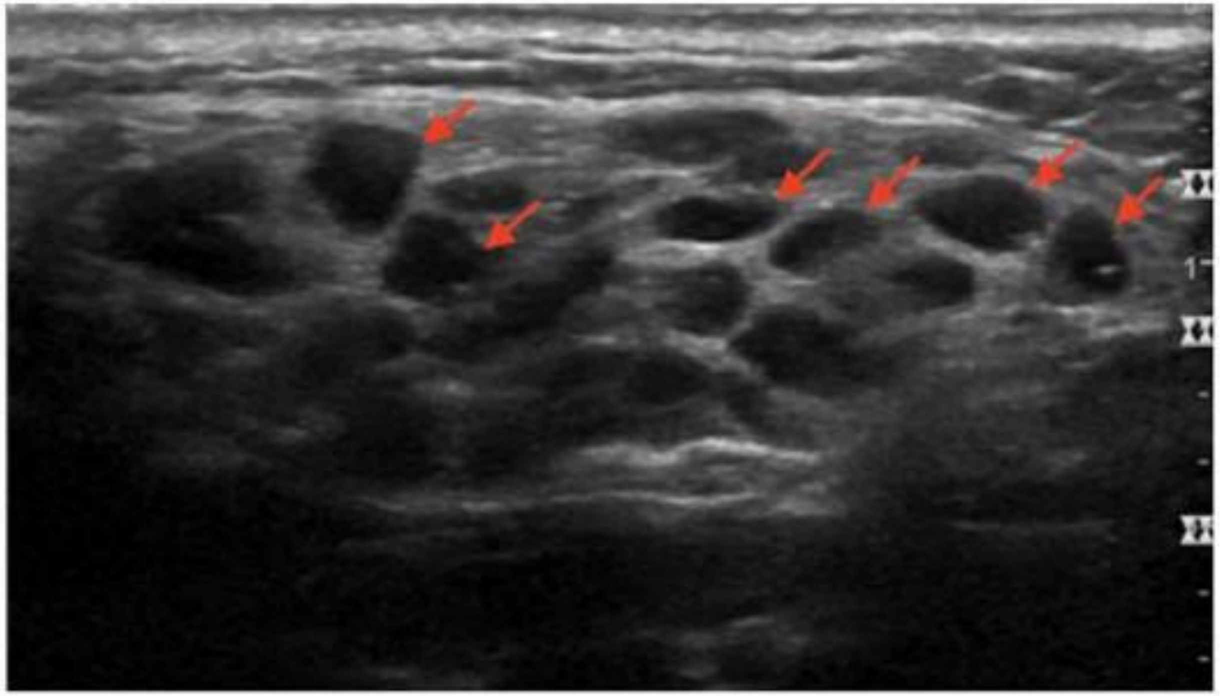


Rheumatology (Oxford) 2022, 61(8), 3341–3350.

pSS



Lymphoma



RMD Open. 2021 Apr;7:e001516.

Open Access Rheumatol. 2022 Sep 1:14:147-160.

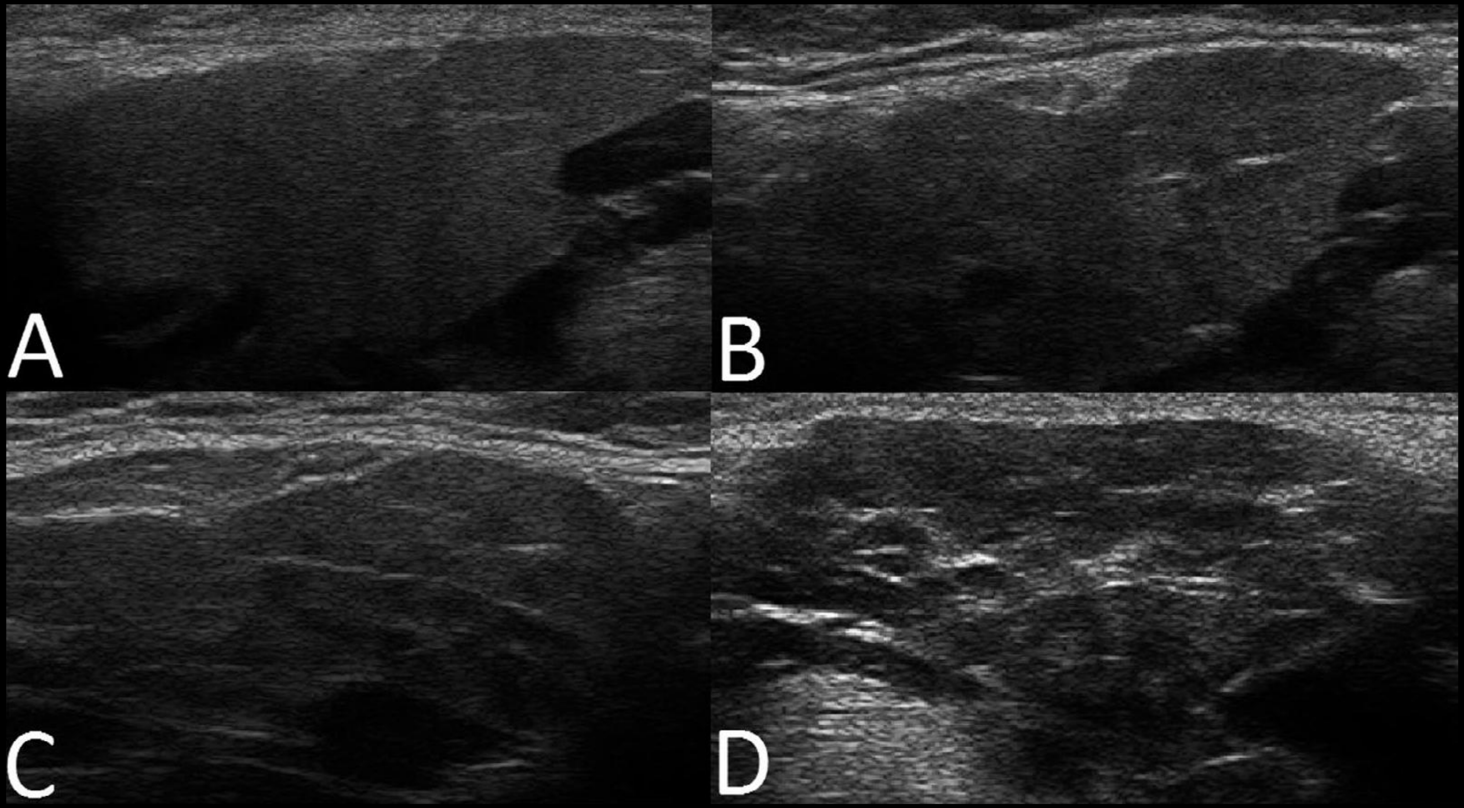
Salivary glands ultrasonography (SGUS)

- Interest regarding the major salivary glands ultrasonography examination (SGUS) as a diagnostic tool for primary SS is increasing.

- **SGUS**
  - A supplement, or even alternative, to minor salivary gland biopsy
  - A noninvasive, diagnostic method
  - Further studies on juvenile SS are needed

Arthritis Care Res (Hoboken). 2020 Jan;72(1):78-87.

Representative major salivary gland ultrasonography images of submandibular glands. A, grade 0; B, grade 1; C, grade 2; and D, grade 3, with grades 0–1 corresponding to normal-appearing morphology, and grades 2–3 corresponding to pathologic changes.



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